

Enrolment Number										
------------------	--	--	--	--	--	--	--	--	--	--

Total No. of printed pages = 04

Monsoon, 2025

B. Tech 5th Semester Examination

ENGINEERING ECONOMICS, ESTIMATION & COSTING

Course Code: BCE23503T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

PART – A

1. Answer the following questions:

1x5=5

- a) The Demand curve is always _____
 - i) Downward sloping
 - ii) Upward rising
 - iii) U-shaped
 - iv) L-Shaped
- b) Index numbers measures the changes in _____
 - i) Price
 - ii) Income
 - iii) Employment
 - iv) All of the above
- c) NNP is _____
 - i) GNP-depreciation
 - ii) GNP-Factor income from abroad
 - iii) GNP-transfer payment
 - iv) GNP-tax
- d) In oligopoly market, number of firms are _____
 - i) Large
 - ii) Few
 - iii) Many
 - iv) One
- e) Example of monopoly include-
 - i) Google
 - ii) Microsoft
 - iii) Both of the above
 - iv) None of the above

2. Answer any *two*:

2x10=20

- a) Discuss the determination of price and output under Monopoly with the help of appropriate diagrams. (10)
- b) What is GST? What are the different types of GST? Explain the advantages of GST. (2+4+4=10)

c) What are public goods? Write the features of public goods. Explain why existence of public goods leads to market failure. (2+4+4=10)

✓ d) Discuss the steps in the construction of index number. Find the price index in 2000, from the following data: (6+4=10)

Commodity	Weight (W)	Price in 1990	Price in 2000
A	6	20	25
B	4	5	10
C	2	15	30
D	4	25	30
E	10	200	450

PART - B

Question No. 6 is mandatory. Attempt any two (02) from the rest.

6. a) The main purpose of preparing an estimate is to: (1)
- Calculate the load on structure
 - Determine the cost of the work\
 - Design the structure
 - Determine the strength of materials
- b) The contingency amount in an estimate is usually: (1)
- 0.5% to 1% of the total estimate
 - 2% to 5% of the total estimate
 - 10% to 15% of the total estimate
 - 20% of the total estimate
- c) The most accurate estimate is: (1)
- Preliminary estimate
 - Plinth area estimate
 - Detailed estimate
 - Approximate estimate
- d) Security deposit is generally: (1)
- 1% of contract value
 - 2% to 10% of contract value
 - 20% of contract value
 - 0.5% of contract value
- e) A contract document does not include: (1)
- General conditions of contract
 - Detailed estimate
 - Drawings and specifications
 - Tender form and acceptance letter

7. A) Critically discuss different approximate methods of estimating. (5)

B) Prepare a preliminary estimate of a four-storied office building having total carpet area of 2700 sq.m for obtaining the administrative approval of the government, given the following data. It may be assumed that 30% of the built-up area will be taken up by corridors, verandah, lavatories, staircase etc. Assume following details a) Plinth area rate is Rs. 1850/- per sq.m, b) Special architectural

treatment requiring additional 1.5% of building cost. c) Deeper foundation at site requiring additional 2% of building cost. d) Water supply and sanitary installation requiring additional 8% of building cost. e) Internal electrical installation requiring additional 10% of building cost. f) Other services requiring additional 6% of building cost. g) Contingencies - 3.5%, h) Supervision charges - 10 %.

(5)

Or,

Estimate the quantity of reinforcement required for a beam, shown in figure 1 below. Assume bearing width as 300 mm, depth of beam 400 mm, width of beam as 300 mm and clear cover as 40 mm. Assume any other data, if required.

(5)

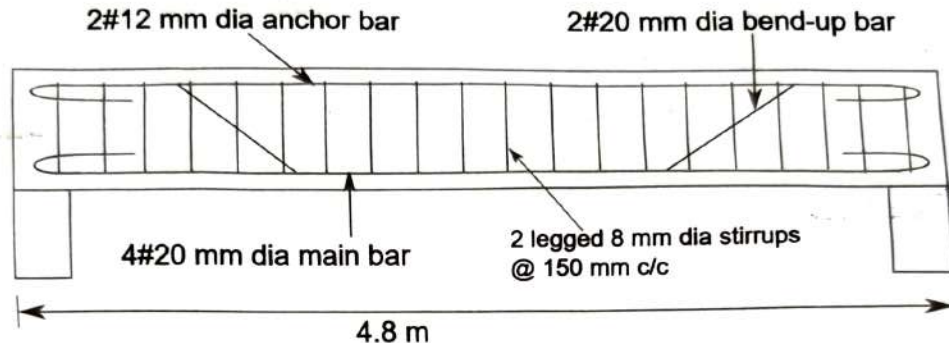


Figure 1

8. Figure 2 shows the plan and sectional elevation of a single room quarter. Prepare the detailed estimate of quantities for the following items with respective general specifications.

- Earthwork in excavation,
- Foundation and plinth with first class brickwork in 1:6 cement and local sand mortar over lime concrete,
- DPC (2.5 cm thick) with 1:2:4 cement concrete,
- Superstructure masonry using first class brickwork in 1:6 cement and local sand mortar.

(10)

9. Estimate the quantity of earthwork required for filling or cutting for a portion of road from the following data – Formation width of road = 12 m, side slope 1.5:1 in banking and cutting. RLs of the ground and formation are as follows:

Distance (m)	0	100	200	300	400	500	600	700	800	900
RL of Ground (m)	114.5	114.75	115.25	115.20	116.10	116.85	117.40	118.25	118.10	117.75
RL of Road (m)	115.0	Upward gradient of 1 in 200 up to 600 m						Downward gradient of 1 in 400 after 600 m		

Assume any other data if required.

(10)

10. A) Summarize the requirements of any contract.

(4)

B) List different types of tenders.

(4)

C) Differentiate between the terms “Earnest Money” and “Security Deposit” used for any contract.

(2)

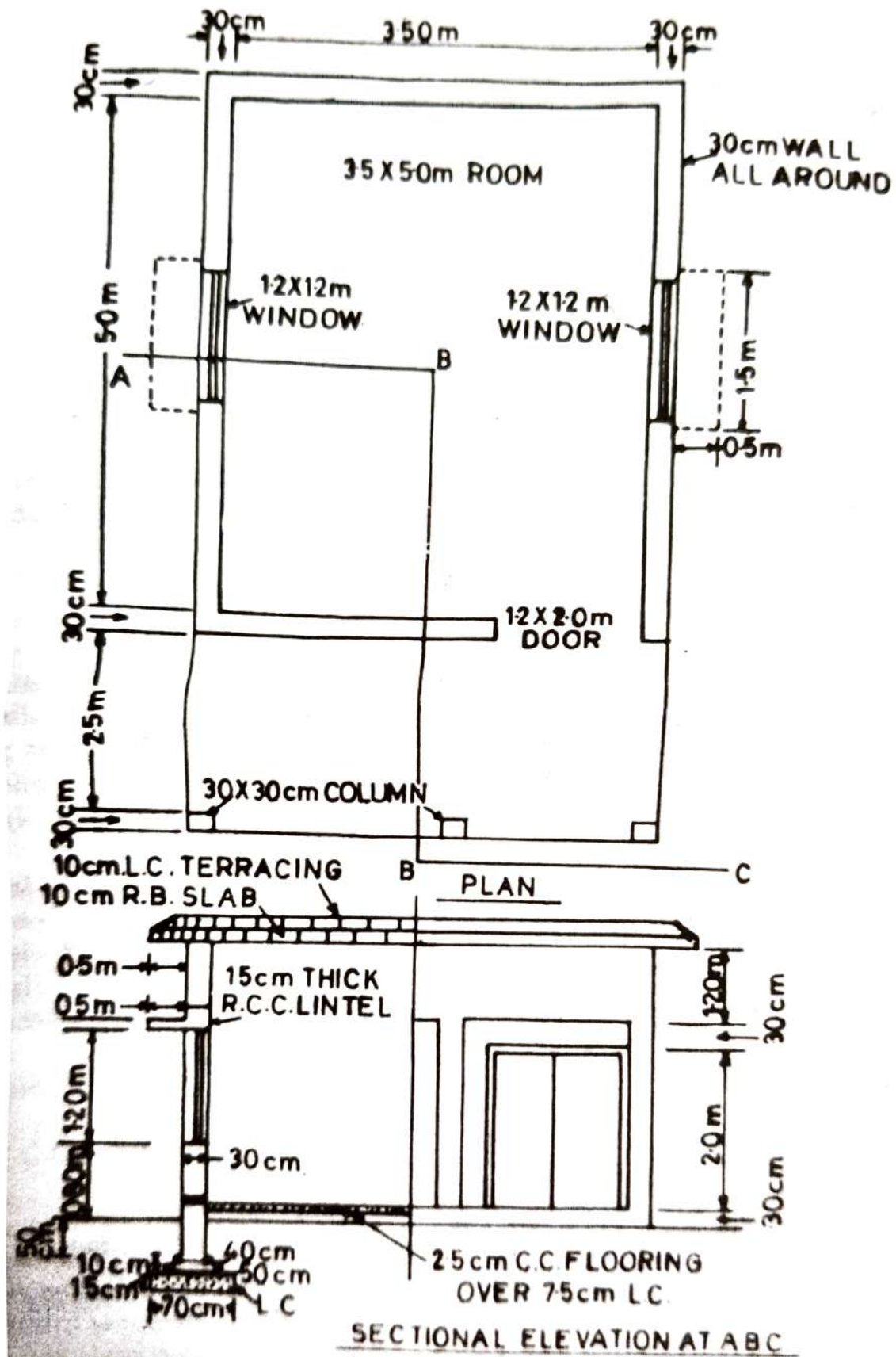


Figure 2